

NIT Delhi, Bachelor of Technology in Electronics and Communication Engineering	Delhi, India
CGPA 8.33	August 2023 – June 2027
IIT Madras, Foundation in Data Science and Programming	Madras, India
CGPA 8.25	June 2023 – September 2024
IIT Madras, Diploma in Data Science	Madras, India
PROJECT CGPA 9.0	October 2024 – December 2025

SKILLS & INTERESTS

- **Languages:** Python, C++, C
- **Core CS:** DSA, OOP, DBMS, Operating Systems, Computer Networks, SQL
- **AI/ML:** Machine Learning, Deep Learning, Computer Vision, Semantic Segmentation, Time Series Forecasting, LiDAR, Sensor Fusion, 3D Perception
- **Libraries & Frameworks:** PyTorch, scikit-learn, OpenCV, Pandas, NumPy, Hugging Face, Qiskit
- **ML Algorithms:** Linear/Logistic Regression, Decision Trees, Random Forest, XGBoost, LightGBM, Gradient Boosting, SVM, KNN, Naive Bayes, K-Means, PCA
- **Tools:** Git, GitHub, Flask, FastAPI, Streamlit, GCP, TFLite, Postman, Jupyter, VS Code
- **Professional Skills:** Leadership, Communication, Problem Solving, Teamwork

EXPERIENCE

Defence Research and Development Organisation (DRDO) – Centre for Artificial Intelligence and Robotics (CAIR)

- AI/ML and Computer Vision InternMay 2026 – Present
- Engineered an end-to-end railway perception pipeline for semantic segmentation and Camera-LiDAR obstacle detection using PyTorch and OpenCV on **RailSem19 (8,500+ images)** and the **OSDaR23** dataset.
 - Benchmarked U-Net, FCN, PSPNet, DeepLabV3+, and SegFormer, achieving **99.47% IoU** (background) and **92% IoU** (railway track).
 - Built preprocessing, augmentation, training, validation, checkpointing, and inference pipelines for **8,500+ annotated railway images**.
 - Developed Camera-LiDAR fusion using camera calibration to process **197K+ LiDAR points per frame** for 3D obstacle localization.
 - Implemented railway track extraction using skeletonization, graph traversal, connected components, and morphological filtering.

Defence Research and Development Organisation (DRDO) – Scientific Analysis Group (SAG)

- AI/ML and Quantum Machine Learning InternMay 2025 – August 2025GitHub
- Developed and deployed a Flask-based ML web application for Bangalore house price prediction, achieving **84.5% accuracy**.
 - Benchmarked Linear Regression, Decision Tree, Random Forest, and Gradient Boosting to identify the best-performing regression model.
 - Built an end-to-end ML pipeline for preprocessing, feature engineering, evaluation, visualization, and model serialization.
 - Implemented Quantum Machine Learning workflows in Qiskit using quantum feature maps and variational quantum classifiers.

PROJECTS

Intelligent Potato Disease Detection System (Deep Learning, GCP)GitHub

- Developed a CNN-based potato disease detection system using the PlantVillage dataset, achieving **99.78% accuracy**; optimized inference with TensorFlow Lite (TFLite).
- Built and deployed a FastAPI backend on GCP with ReactJS and React Native interfaces for real-time disease prediction.

Business Data Management Capstone (IIT Madras)GitHub

- Built a business analytics solution for a B2B manufacturer, identifying **66% delayed payments**, **77 excess workers**, and **2× higher sourcing delays**.
- Delivered data-driven recommendations to improve financial stability, workforce utilization, and supply chain efficiency.

Audience Demand Forecasting System – Time-Series ML (IIT Madras)GitHub

- Built a time-series forecasting pipeline on **100K+ records**, engineering **15+ temporal features** from booking and POS data.
- Trained Random Forest, XGBoost, and LightGBM models, achieving **~0.85 R²** with time-aware validation.

Robotics & Assistive Technology

- Arduino-based spider robot for automated grass cutting, pesticide spraying, smart blind stick with ultrasonic obstacle detection.

POSITION OF RESPONSIBILITY & ACHIEVEMENTS

- **Academic:** 96.37 percentile in JEE Main 2023; Top 0.2% in CBSE Class X; SOF IMO & NSO awards.
- **Leadership & Responsibilities:** Captain, ECE Badminton Team (ZEAL 2024, 2025); Sports Coordinator; Managed T&P database; Represented NIT Delhi at Inter NIT (MST 2024, 2025; ITUSA 2024).
- **Achievements:** SIH 2025 Participant; Tech Odyssey Participant; IIT Madras badges in Statistics, Mathematics, and English.

CERTIFICATIONS

- Google AI Essentials – Google
- Dec 2024